

WORKFORCE MANAGEMENT SYSTEM: ATTENDANCE, SHIFT, AND LEAVE TRACKING

Faculty

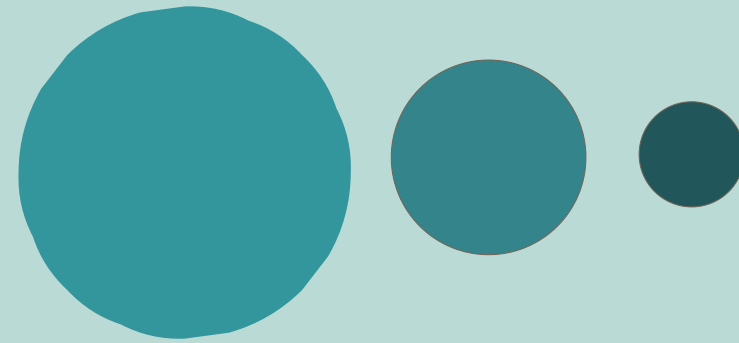
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AIML WHALES

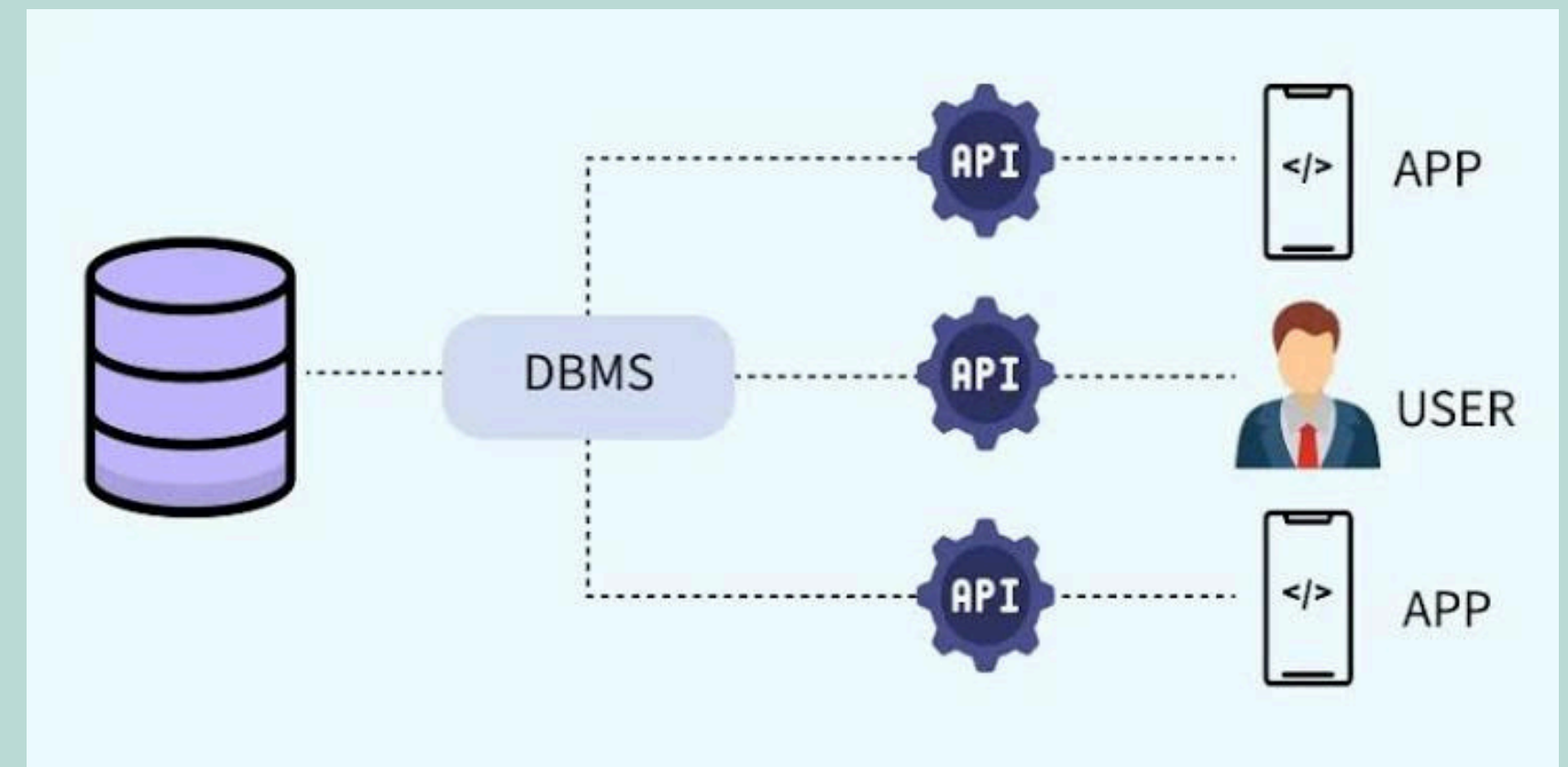
Overview



Design and Implementation of a Database Management System

The Workforce Management System is a centralized relational database designed to automate and streamline core HR operations, specifically focusing on employee attendance, shift scheduling, and leave management. By replacing fragmented manual tracking methods with a unified architecture, the system ensures data integrity and enforces critical business rules, such as validating daily work hours and managing leave approval statuses.

Ultimately, this robust database provides a single, reliable source of truth that simplifies administrative workflows, optimizes workforce coverage, and enhances overall organizational efficiency.



CORE PROBLEM

Current Challenge:

Many organizations still rely on manual ledgers, spreadsheets, or disjointed software to track employee attendance, shift schedules, and leave requests. This leads to data redundancy, payroll calculation errors, difficulty in tracking leave balances, and poor visibility into shift coverage.

Proposed Solution:

This project aims to design and implement a centralized, relational Database Management System (DBMS) that automates the tracking of workforce attendance. The system will ensure data integrity, streamline leave approvals, enforce scheduling rules, and provide a single source of truth for HR reporting.

Employee Database Management

Add Employee

Employee List

Thomas Leannon

X

Faye Sauer

X

Deven Halvorson

X

Melisa Schuppe

X

Dell Kris

X

Marcia Gutmann

X

Jarrold Ortiz

X

Gabriella Wilkinson

X

Elisabeth Hayes

X

Jaime Reichel

X

Employee Information

THOMAS LEANNON (43)

Address1

Thomas.Leannon@dummyapis.com

Mobile - 4121091095

DOB - 26/08/1979



OPERATIONAL CONSTRAINTS AND SYSTEM LOGIC

- 1 An employee can belong to only one department at a time, but a department can have many employees.
- 2 An employee can only have one active clock-in session per day.
- 3 A clock-out time must always be chronologically after the clock-in time.
- 4 A leave request must have a status of 'Pending', 'Approved', or 'Rejected' (default is 'Pending').
- 5 Two overlapping leave requests cannot be approved for the same employee.

CORE ENTITIES

Employee

The core entity.

Stores: Employee ID, Name, Email, Phone, Hire Date, Department ID.

Department

Groups employees together.

Stores: Department ID, Department Name, Manager ID

Shift

Defines the working hours.

Stores: Shift ID, Shift Name, Start Time, End Time.

Attendance

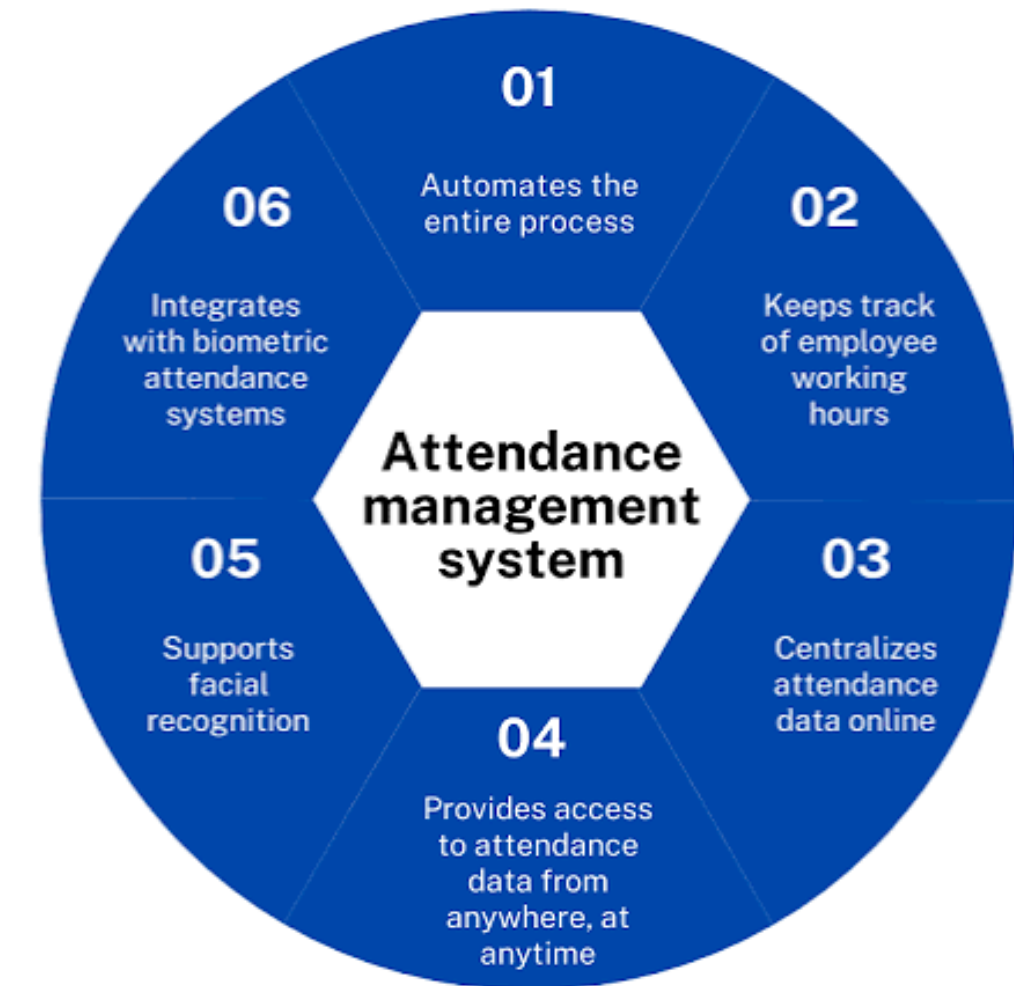
The daily log of when employees actually work.

Stores: Log ID, Employee ID, Date, Clock-In Time, Clock-Out Time, Status.

Leave Request

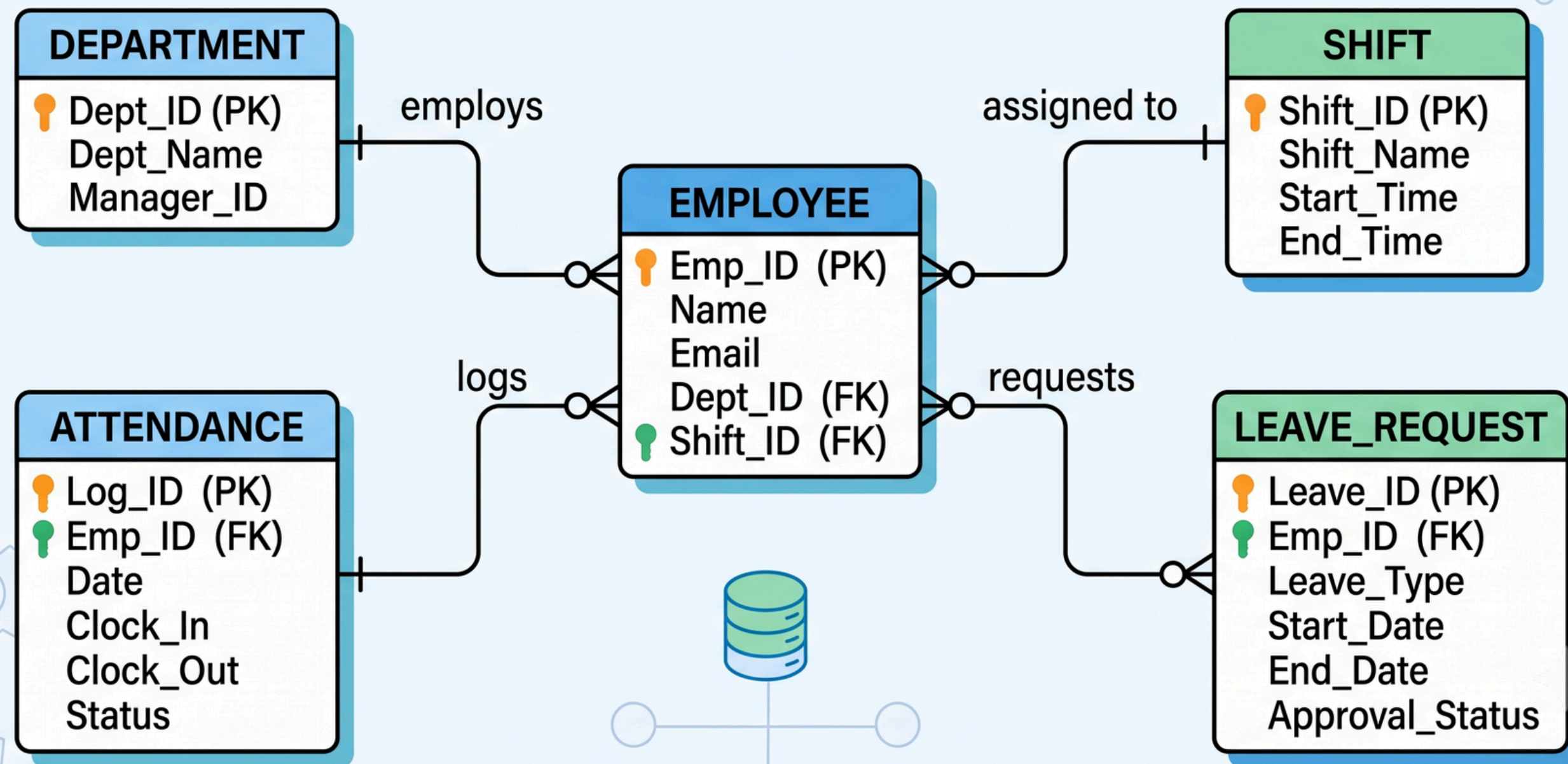
Tracks time off.

Stores: Leave ID, Employee ID, Leave Type, Start Date, End Date, Approval Status.





PRIMARY ENTITY-RELATIONSHIP (ER) DIAGRAM: WORKFORCE MANAGEMENT SYSTEM





THANK
YOU

